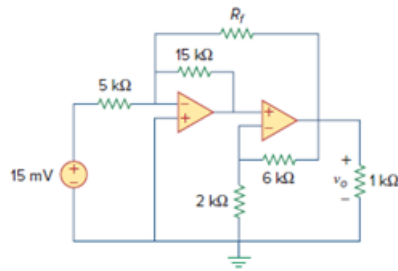


Problem

Find v_o in the circuit of Fig. 5.95, assuming that $R_f = \infty$ (open circuit).

Figure. 5.95:



Step-by-step solution

Step 1 of 4

Draw the circuit diagram as shown in Figure 1.

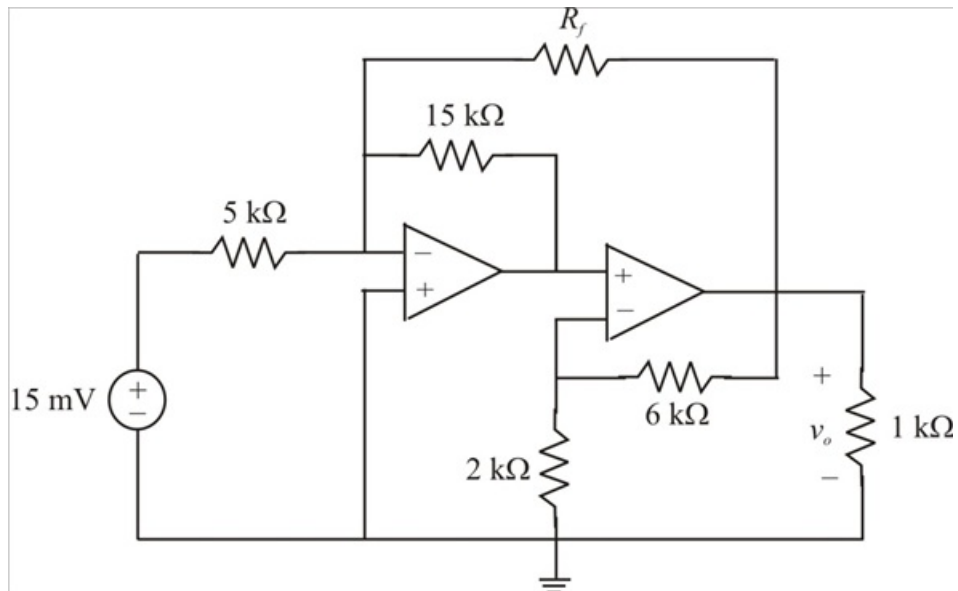


Figure 1

Step 2 of 4

The circuit is labeled as shown in Figure 2.

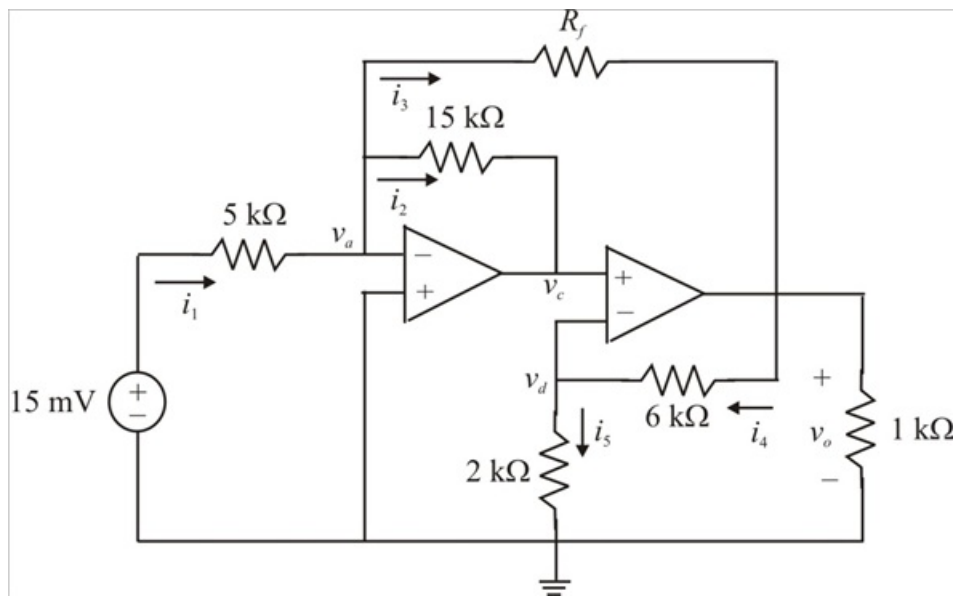


Figure 2

Step 3 of 4

Assuming that $R_f = \infty$ (open circuit)

Applying Kirchhoff's current law to node v_a ,

$$i_1 = i_2 + i_3$$

$$\frac{(15 \times 10^{-3}) - v_a}{5} = \frac{v_a - v_c}{15} + \frac{v_a - v_o}{\infty}$$

$$(45 \times 10^{-3}) - 3v_a = v_a - v_c + 0$$

$$4v_a - v_c = 45 \times 10^{-3}$$

Therefore $4v_a - v_c = 45 \times 10^{-3}$ (1)

But $v_a = 0$ since non-inverting terminal is grounded. Then equation (1) becomes,

$$4v_a - v_c = 45 \times 10^{-3}$$

$$4(0) - v_c = 45 \times 10^{-3}$$

$$v_c = -45 \text{ mV}$$

Therefore $v_c = -45 \text{ mV}$ (2)

Step 4 of 4

Applying Kirchhoff's current law to node v_d ,

$$i_4 = i_5$$

$$\frac{v_o - v_d}{6} = \frac{v_d - 0}{2}$$

$$v_o - v_d = 3v_d$$

$$v_o = 4v_d$$

Therefore $v_o = 4v_d$ (3)

But $v_c = v_d$

Then equation (3) becomes,

$$v_o = 4v_d$$

$$= 4v_c$$

$$= 4(-45 \times 10^{-3})$$

$$= -180 \text{ mV}$$

Therefore the output voltage v_o is $\boxed{-180 \text{ mV}}$.